SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)
Assessment Tool : Short Answer Test
Weightage: 10%

QUESTIONS: **Total Marks: 20**

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I VARSHA S(192462003) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Internet Programming | **COURSE CODE:** CSA43 | **COURSE OUTCOME:** CO-2

Internet Programming

CO-2: Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior.

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5 Marks

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Answer

Responsive Web Design is an approach to web development in which a website automatically adjusts its layout, content, images, and elements according to the screen size and device being used.

It is important because users access websites using mobile phones, tablets, laptops, and desktop computers. A responsive website provides a consistent and user-friendly experience on all these devices.

Key CSS Techniques

- **Media Queries:** Apply different CSS rules for different screen sizes.
- **Flexible Widths:** Use percentages, `max-width`, and relative units instead of fixed widths.
- **Responsive Images:** Use `max-width: 100%` so images do not overflow their containers.
- **Flexbox:** Arranges elements horizontally or vertically and automatically adjusts their alignment.
- **CSS Grid:** Creates flexible rows and columns for complex layouts.

Example

```
.container {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 20px;
}

@media (max-width: 600px) {
    .container {
        grid-template-columns: 1fr;
    }
}
```

In this example, three columns are displayed on larger screens. When the screen width becomes smaller than 600 pixels, the layout changes to a single column.

Improvements

User experience can be improved by using readable fonts, touch-friendly buttons, proper spacing, optimized images, simple navigation, sufficient color contrast, and testing the website on different screen sizes.

Question 2

5 Marks

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Answer

Client-side validation is the process of checking user input in the browser before the data is submitted to the server. It helps identify incorrect or missing information immediately.

The main purpose is to improve user experience, reduce unnecessary server requests, and ensure that the input follows the required format.

JavaScript Event Handling

JavaScript can use events such as `submit`, `click`, and `input` to perform validation. For example, the `submit` event can be used to validate form fields before allowing the form to be submitted.

Regular Expressions

A regular expression, or regex, is a pattern used to check whether input follows a particular format. It can be used to validate email addresses, phone numbers, passwords, and other text patterns.

Example

```
const emailPattern =
    /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

const phonePattern =
    /^[0-9]{10}$/;

if (!emailPattern.test(email)) {
    alert("Enter a valid email");
}

if (!phonePattern.test(phone)) {
    alert("Enter a 10-digit phone number");
}
```

Improvements

- Display clear error messages near the input field.
- Use appropriate input types such as email and tel.
- Provide immediate feedback to the user.
- Use both client-side and server-side validation.
- Never rely only on client-side validation for security.

Question 3

5 Marks

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Answer

The **CSS Box Model** describes how every HTML element is represented as a rectangular box. It consists of four main components:

1. **Content:** The actual text, image, or other content inside an element.
2. **Padding:** The space between the content and the border.
3. **Border:** The visible boundary surrounding the padding and content.
4. **Margin:** The outer space between the element and other elements.

CSS Positioning

CSS provides different positioning techniques such as `static`, `relative`, `absolute`, `fixed`, and `sticky`.

Relative positioning moves an element relative to its normal position. **Absolute** positioning removes an element from the normal document flow and positions it relative to its nearest positioned ancestor.

Causes of Layout Breaking

- Using fixed widths that are too large.
- Excessive margins and padding.
- Images larger than their containers.
- Incorrect use of absolute positioning.
- Not using responsive media queries.
- Content overflowing the screen.

Media Query Example

```css
.container {
    width: 90%;
    margin: auto;
}

@media (max-width: 768px) {
    .container {
        width: 95%;
    }

    .menu {
        flex-direction: column;
    }
}
```

Best Practices

Use flexible layouts, relative units, CSS Flexbox or Grid, responsive images, the `box-sizing: border-box` property, and appropriate media queries. Avoid unnecessary fixed widths and excessive absolute positioning.

Question 4

5 Marks

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Answer

The **Document Object Model (DOM)** is a programming interface that represents an HTML document as a tree structure of objects. Each HTML element, attribute, and piece of text can be accessed through the DOM.

The DOM allows JavaScript to access and modify webpage elements dynamically without requiring the entire page to reload.

DOM Manipulation

JavaScript can select HTML elements using methods such as `getElementById()`, `querySelector()`, and `querySelectorAll()`. The selected elements can then be modified using properties such as `textContent`, `innerHTML`, `style`, and class manipulation.

Example

Content Updated Successfully!

Update Content

```
function updateContent() {
    document.getElementById("dynamicText")
        .textContent = "Content Updated!";
}
```

When the button is clicked, the JavaScript function selects the HTML element using its ID and changes its text. The page does not need to be reloaded.

Improvements

- Update only the required DOM elements.
- Avoid unnecessary DOM manipulations.
- Use event listeners instead of excessive inline events.
- Use efficient selectors.
- Provide visual feedback during dynamic updates.
- Keep the interface simple and responsive.